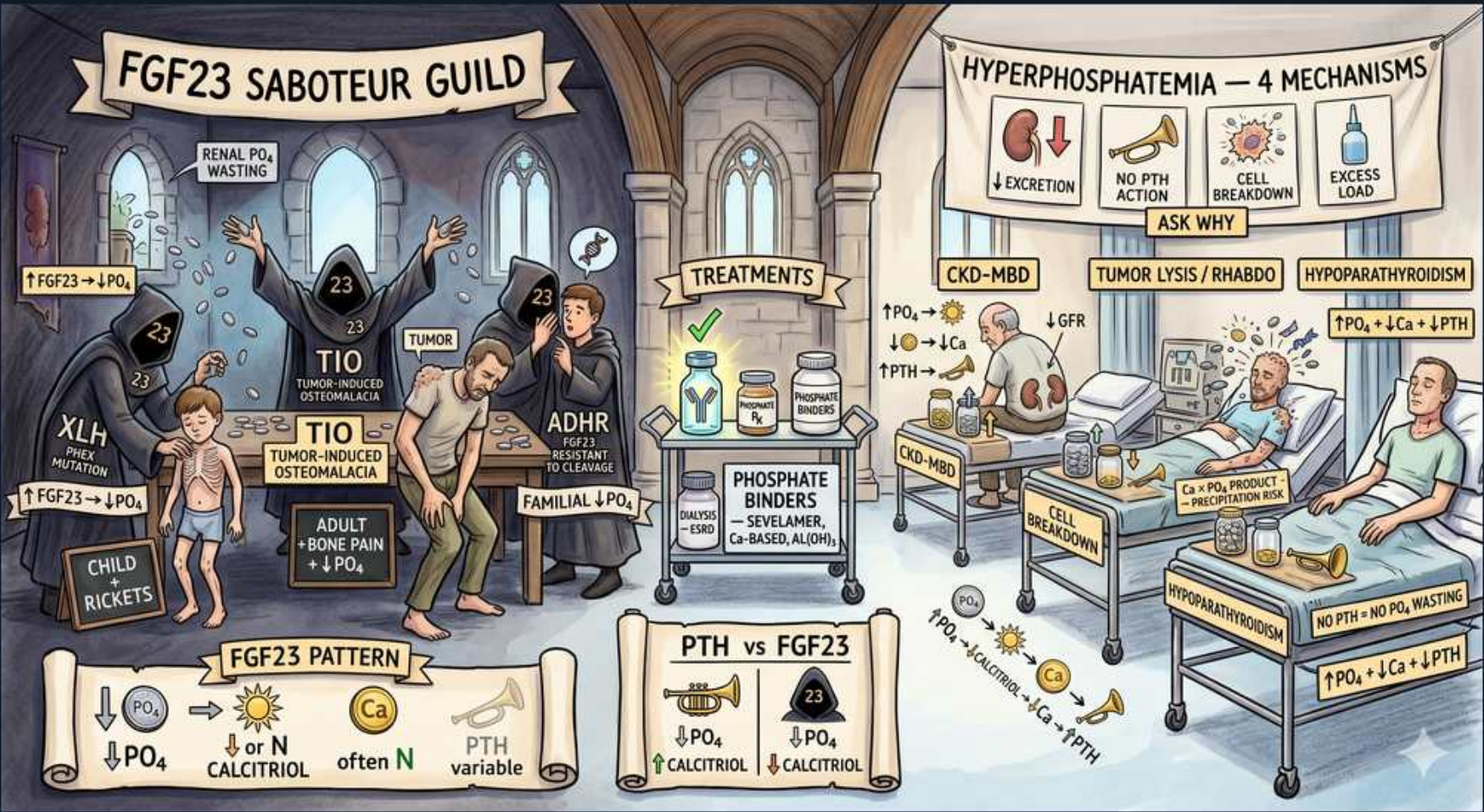


Calcium/PTH/VitD — The FGF23 Saboteur Guild & Hyperphosphatemia Ward



1. FGF23 saboteur guild

WHAT YOU SEE

Banner over hooded saboteurs labeled FGF23 Saboteur Guild causing renal PO4 wasting.

WHAT IT ENCODES

FGF23 (fibroblast growth factor 23) is a bone-derived phosphatonin secreted by osteocytes/osteoblasts; it acts on the proximal tubule via the FGFR1–Klotho co-receptor to downregulate the NaPi-2a/2c sodium-phosphate cotransporters, producing renal phosphate wasting and hypophosphatemia. Critically, it also inhibits renal 1-alpha-hydroxylase (CYP27B1) and stimulates 24-hydroxylase (CYP24A1), so it LOWERS calcitriol (1,25-vitamin D). The net 'FGF23 syndrome' signature is low PO4 + inappropriately low/normal calcitriol. Mnemonic: 'FGF23 = Phosphate Goes, Free of vitamin D' — it wastes phosphate and suppresses active D.

BOARD TRAP

WRONG ANSWER: 'FGF23 excess raises calcitriol because it's a hormone responding to low phosphate.' Discriminator: FGF23 LOWERS phosphate AND LOWERS calcitriol — the opposite of the appropriate physiologic response (low PO4 should normally stimulate 1-alpha-hydroxylase). This dissociation (low PO4 with non-elevated calcitriol) is the fingerprint of an FGF23-mediated disorder versus dietary phosphate loss.

2. XLH child with rickets

WHAT YOU SEE

A child figure labeled XLH (PHEX mutation) with bowed legs and rickets.

WHAT IT ENCODES

X-linked hypophosphatemia (XLH) is the most common heritable form of rickets, caused by a loss-of-function PHEX gene mutation (X-linked dominant). Loss of PHEX leads to increased circulating intact FGF23, driving renal phosphate wasting, hypophosphatemia, and impaired bone mineralization with bowing, short stature, and dental abscesses. Labs: low PO4, normal/low calcitriol, normal calcium and PTH, elevated ALP. Treatment has shifted from oral phosphate + calcitriol to burosumab (anti-FGF23 monoclonal antibody), which is now first-line for XLH.

BOARD TRAP

WRONG ANSWER: 'Give high-dose vitamin D / cholecalciferol to cure the rickets.' Discriminator: XLH rickets is driven by phosphate wasting, NOT vitamin D deficiency — 25-OH-D is normal and calcitriol is low/normal as a CONSEQUENCE of FGF23, not the primary defect. Plain vitamin D won't fix it; you need phosphate repletion or burosumab to neutralize FGF23.

3. TIO tumor

WHAT YOU SEE

A figure with a tumor labeled TIO (tumor-induced osteomalacia), down PO4 and bone pain.

WHAT IT ENCODES

Tumor-induced osteomalacia (TIO, oncogenic osteomalacia) is an acquired paraneoplastic syndrome in which a small, often occult mesenchymal tumor (phosphaturic mesenchymal tumor) secretes FGF23, producing the same low-PO4, low-calcitriol picture as XLH but in an adult. Classic presentation: insidious bone pain, proximal muscle weakness, and fractures with persistent unexplained hypophosphatemia. Tumors are typically tiny and hide in bone or soft tissue (feet, head/neck); localization often requires 68-Ga-DOTATATE PET or octreotide scanning. Resection is curative.

BOARD TRAP

WRONG ANSWER: 'Treat the chronic hypophosphatemic osteomalacia with lifelong phosphate/calcitriol and stop looking.' Discriminator: TIO is potentially CURABLE by finding and resecting the FGF23-secreting tumor — an adult with new unexplained hypophosphatemic osteomalacia warrants a tumor hunt (DOTATATE PET), not just symptomatic repletion. Missing the tumor is the tested error.

4. ADHR resistant to cleavage

WHAT YOU SEE

Figure labeled ADHR with FGF23 resistant to cleavage, familial low PO4.

WHAT IT ENCODES

Autosomal dominant hypophosphatemic rickets (ADHR) is caused by a gain-of-function mutation in the FGF23 gene itself at the RXXR furin cleavage site, rendering FGF23 resistant to proteolytic inactivation so it accumulates. Phenotype overlaps XLH (low PO4, low/normal calcitriol, phosphaturia) but inheritance is autosomal dominant and onset is variable — penetrance and severity fluctuate with iron status (iron deficiency raises FGF23, worsening disease). Mnemonic: ADHR = FGF23 that 'won't die' (uncleavable).

BOARD TRAP

WRONG ANSWER: 'ADHR is due to overproduction of FGF23 from a tumor.' Discriminator: ADHR is a germline FGF23 mutation that blocks DEGRADATION (cleavage-resistant), not increased synthesis or a tumor (that's TIO). Iron deficiency unmasking/worsening symptoms is a uniquely ADHR clue.

5. FGF23 pattern scroll

WHAT YOU SEE

Bottom scroll: down PO4, down or normal calcitriol.

WHAT IT ENCODES

The unifying FGF23-excess lab pattern: LOW serum phosphate with INAPPROPRIATELY low or normal calcitriol (1,25-D), normal serum calcium, normal/normal-ish PTH, and elevated urinary phosphate (high fractional excretion of phosphate, FEPO4 >5–20%). The 'tell' is that calcitriol fails to rise despite hypophosphatemia, because FGF23 directly suppresses 1-alpha-hydroxylase. ALP is often elevated reflecting active osteomalacia.

BOARD TRAP

WRONG ANSWER: 'Low phosphate always means dietary insufficiency or a GI cause — calcitriol should be high.' Discriminator: in FGF23 excess, calcitriol does NOT rise the way it should with low PO4, and urinary phosphate is inappropriately HIGH (renal wasting). Checking FEPO4 and seeing renal loss distinguishes FGF23/renal causes from dietary/GI loss (where urine PO4 is appropriately low).

6. PTH vs FGF23 scroll

WHAT YOU SEE

Comparison: PTH lowers PO4 and raises calcitriol; FGF23 lowers PO4 and lowers calcitriol.

WHAT IT ENCODES

Both PTH and FGF23 are phosphaturic (both downregulate proximal-tubule NaPi cotransporters and lower serum phosphate), but they have OPPOSITE effects on vitamin D: PTH STIMULATES 1-alpha-hydroxylase (raises calcitriol), whereas FGF23 INHIBITS it (lowers calcitriol). They also differ on calcium: PTH raises serum Ca; FGF23 has little direct Ca effect. FGF23 and PTH form a feedback loop — phosphate and calcitriol stimulate FGF23, and FGF23 in turn suppresses PTH. Mnemonic: 'PTH pumps up D, FGF23 fights D.'

BOARD TRAP

WRONG ANSWER: 'PTH and FGF23 do the same thing, so calcitriol behaves identically in hyperparathyroidism and FGF23 excess.' Discriminator: the calcitriol direction splits them — high PTH → high calcitriol (low PO4, high Ca, high calcitriol = primary HPT picture), while high FGF23 → low calcitriol (low PO4, normal Ca, low calcitriol). Calcitriol level is the single best discriminator.

7. Phosphate binders treatment

WHAT YOU SEE

Central treatment shelf with binders (sevelamer, Ca-based, aluminum) and burosumab.

WHAT IT ENCODES

Two opposite problems, two opposite treatments. HYPERphosphatemia (CKD-MBD): dietary phosphate restriction plus oral phosphate binders taken WITH meals — sevelamer (non-calcium, preferred to avoid Ca load/vascular calcification), calcium acetate/carbonate (also treats hypocalcemia but risks calcification), lanthanum, and short-term aluminum hydroxide (avoid chronic use due to aluminum toxicity/encephalopathy/osteomalacia). FGF23-driven HYPOphosphatemia (XLH/TIO): burosumab, an anti-FGF23 monoclonal antibody, now first-line.

BOARD TRAP

WRONG ANSWER: 'Aluminum hydroxide is a fine long-term phosphate binder in CKD.' Discriminator: chronic aluminum binders cause aluminum accumulation → adynamic bone disease/osteomalacia, microcytic anemia, and dialysis encephalopathy. Non-calcium binders (sevelamer) are preferred long-term; aluminum is only for short-term severe hyperphosphatemia.

8. Hyperphosphatemia 4 mechanisms

WHAT YOU SEE

Top-right banner: decreased excretion, no PTH action, cell breakdown, excess load.

WHAT IT ENCODES

Four buckets for hyperphosphatemia: (1) DECREASED renal excretion — chronic/acute kidney disease is by far the most common cause (kidney can't dump PO₄ once GFR falls below ~25–30); (2) LOW PTH action — hypoparathyroidism or pseudohypoparathyroidism removes the phosphaturic drive; (3) CELLULAR RELEASE — tumor lysis syndrome, rhabdomyolysis, hemolysis dump intracellular phosphate; (4) EXCESS LOAD — phosphate-containing enemas/laxatives (sodium phosphate), IV phosphate, or massive vitamin D intake. Spurious hyperphosphatemia (pseudohyperphosphatemia) can occur with hemolyzed samples or paraproteinemia.

BOARD TRAP

WRONG ANSWER: 'A sodium-phosphate bowel prep is harmless regarding phosphate.' Discriminator: oral sodium phosphate purgatives can cause acute severe hyperphosphatemia and acute phosphate nephropathy (nephrocalcinosis), especially in elderly/CKD patients — an 'excess load' cause that's a classic exam vignette.

9. CKD-MBD bed

WHAT YOU SEE

Patient labeled CKD-MBD with down GFR, up PO₄, up PTH, low calcitriol.

WHAT IT ENCODES

CKD–mineral and bone disorder (CKD-MBD): as GFR falls, the kidney can neither excrete phosphate nor make calcitriol. Early, FGF23 rises (first compensatory change) to promote phosphaturia; eventually phosphate retention wins. Low calcitriol + hyperphosphatemia + the resulting hypocalcemia all drive secondary hyperparathyroidism (high PTH). Sequence to remember: ↑FGF23 → ↓calcitriol → ↓Ca + ↑PO₄ → ↑PTH. Long-standing stimulation can become autonomous (tertiary HPT with high Ca). Management targets phosphate (binders), calcitriol/analogs, and calcimimetics (cinacalcet).

BOARD TRAP

WRONG ANSWER: 'CKD-MBD looks like an FGF23-wasting syndrome with low phosphate.' Discriminator: in CKD phosphate is HIGH (retention from low GFR) and calcitriol is LOW — the REVERSE of XLH/TIO where phosphate is low. The shared feature is low calcitriol; the splitter is the phosphate direction (high in CKD, low in FGF23-wasting).

10. Tumor lysis / rhabdo

WHAT YOU SEE

Cell breakdown panel: tumor lysis / rhabdomyolysis releasing PO₄.

WHAT IT ENCODES

Massive cell breakdown releases intracellular phosphate (cells are phosphate-rich), causing acute hyperphosphatemia. Tumor lysis syndrome (after chemotherapy of bulky/high-turnover tumors — Burkitt lymphoma, ALL) gives the tetrad: ↑K, ↑PO₄, ↑uric acid, ↓Ca, often with AKI. Rhabdomyolysis (crush injury, statins, exertion) releases PO₄, K, myoglobin, and CK. Prevention/treatment: aggressive IV hydration, rasburicase/allopurinol for urate, and managing electrolytes.

BOARD TRAP

WRONG ANSWER: 'Give calcium aggressively for the hypocalcemia of tumor lysis.' Discriminator: in tumor lysis the hypocalcemia is driven by HIGH phosphate (Ca-PO₄ precipitation); giving calcium when phosphate is high risks metastatic calcification. Only treat hypocalcemia if symptomatic — first lower phosphate. Replacing PO₄ here would be the opposite error since PO₄ is already high.

11. Cell breakdown bed

WHAT YOU SEE

Patient with arrows: up Ca x PO₄ product, precipitation risk.

WHAT IT ENCODES

When the calcium × phosphate product rises (classically a product >55 mg²/dL²), calcium-phosphate precipitates in soft tissues, causing metastatic/ectopic calcification — vascular calcification, calciphylaxis (in CKD/dialysis), conjunctival/corneal deposits, and nephrocalcinosis. This is the unifying danger of sustained hyperphosphatemia and a major driver of cardiovascular mortality in CKD. It's why phosphate control (binders, dialysis) matters so much.

BOARD TRAP

WRONG ANSWER: 'Soft-tissue/vascular calcification reflects high calcium alone.' Discriminator: it's the Ca × PO₄ PRODUCT, not calcium alone — hyperphosphatemia with even normal calcium can drive calcification (calciphylaxis in dialysis patients is a feared example). Controlling phosphate, not just calcium, is the key intervention.

12. Hypoparathyroidism bed

WHAT YOU SEE

Patient labeled hypoparathyroidism with no PTH/no PO₄ wasting, up PO₄, low Ca, low PTH.

WHAT IT ENCODES

Hypoparathyroidism removes PTH's phosphaturic action, so phosphate is retained → hyperphosphatemia, alongside hypocalcemia (no PTH to mobilize Ca or activate calcitriol) and a LOW/inappropriately normal PTH. Causes: post-surgical (thyroidectomy/parathyroidectomy — most common), autoimmune (APS-1), DiGeorge (22q11 deletion), and hypomagnesemia (impairs PTH secretion/action). Symptoms are those of hypocalcemia: perioral numbness, Chvostek/Trousseau signs, tetany, QT prolongation. Pseudohypoparathyroidism gives the same Ca/PO₄ picture but with HIGH PTH (end-organ resistance).

BOARD TRAP

WRONG ANSWER: 'High phosphate + low calcium means CKD/secondary hyperparathyroidism.' Discriminator: check PTH. Hypoparathyroidism = high PO₄, low Ca, LOW PTH. CKD secondary HPT = high PO₄, low Ca, but HIGH PTH. Pseudohypoparathyroidism = high PO₄, low Ca, HIGH PTH (resistance, often with Albright osteodystrophy). The PTH level splits these three.